

Extending symplectic embeddings to symplectic automorphisms

Abstract

Let (V, Ω) be a separable real Hilbert space equipped with a strong symplectic form induced by an orthogonal complex structure $\mathcal{J}$. We prove two extension theorems. First, every bounded symplectic embedding $T : V \rightarrow V$ extends to a symplectic automorphism of a larger symplectic Hilbert space; the construction is elementary and explicit. Second, every commuting pair of *isometric* symplectic embeddings extends to a commuting pair of unitary symplectic automorphisms on a common enlargement. The argument is the symplectic counterpart of Itô's conjugation method for commuting isometries: one dilates the first map by the unilateral defect construction, conjugates the second map along the resulting unitary group, and dilates once more. No double-commutation hypothesis is used.

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1 Setup and notation

Throughout, V is a real separable Hilbert space with inner product g and norm $\|\cdot\|$. Fix $\mathcal{J} \in \mathcal{B}(V)$ satisfying

$$\mathcal{J}^2 = -I, \quad \mathcal{J}^* = -\mathcal{J}. \quad (1)$$

The associated bilinear form

$$\Omega(x, y) := g(\mathcal{J}x, y) \quad (2)$$

is then a *strong* symplectic form: the map $\flat : V \rightarrow V^*$ given by $\flat(x) = \Omega(x, \cdot)$ is a topological isomorphism.

The motivating model is the countable orthogonal sum

$$H = \ell^2(\mathbb{N}; \mathbb{R}^{2d}) = \bigoplus_{n \geq 1} \mathbb{R}^{2d}, \quad \mathcal{J} = \bigoplus_{n \geq 1} J_{2d}, \quad \Omega = \bigoplus_{n \geq 1} \omega_0,$$

where ω_0 is the standard area form on $\mathbb{R}^{2d}$ and J_{2d} is the corresponding orthogonal complex structure.

Definition 1.1. An operator $T \in \mathcal{B}(V)$ is a *symplectic embedding* if

$$\Omega(Tx, Ty) = \Omega(x, y) \quad \text{for all } x, y \in V, \quad (3)$$

equivalently $T^* \mathcal{J} T = \mathcal{J}$. It is a *symplectic automorphism* if it is additionally bijective; in that case $T^{-1} \in \mathcal{B}(V)$ by the bounded inverse theorem, and we write $T \in \text{Sp}(V, \Omega)$.

A symplectic embedding T is called *isometric* if $T^* T = I$ (equivalently, $\|Tx\| = \|x\|$ for every $x \in V$).

Lemma 1.2. (i) $\|\mathcal{J}x\| = \|x\|$ for every $x \in V$.

(ii) Condition (3) holds if and only if $T^* \mathcal{J} T = \mathcal{J}$.

(iii) The form Ω is strong.

Proof. From (1) one has $\mathcal{J}^* \mathcal{J} = I$, so $\|\mathcal{J}x\|^2 = g(x, \mathcal{J}^* \mathcal{J} x) = \|x\|^2$. Expanding the definitions gives $\Omega(Tx, Ty) = g(T^* \mathcal{J} T x, y)$ and $\Omega(x, y) = g(\mathcal{J}x, y)$, which proves (ii). For (iii), note that $\flat = \iota_g \circ \mathcal{J}$ where $\iota_g : V \rightarrow V^*$ is the Riesz map. $\square$

2 Left inverse, closed range, and the symplectic complement

Lemma 2.1. Let T be a symplectic embedding. Then:

- (i) $\|Tx\| \geq \|x\|/\|T\|$ for all $x \in V$; in particular T is injective;
- (ii) $R := \text{ran } T$ is closed;
- (iii) the operator $L := -\mathcal{J} T^* \mathcal{J}$ satisfies $LT = I$ and $\|L\| \leq \|T\|$;
- (iv) $E := TL$ is a bounded projection onto R , and $F := I - E$ satisfies $F|_R = 0$.

Proof. Using Lemma 1.2,

$$\|x\| = \|\mathcal{J}x\| = \|T^* \mathcal{J} T x\| \leq \|T\| \|Tx\|,$$

which is (i). If $Tx_n \rightarrow z$, then (Tx_n) is Cauchy, hence so is (x_n) by (i); if $x_n \rightarrow x$, continuity yields $Tx = z$, so R is closed. Next,

$$LT = (-\mathcal{J} T^* \mathcal{J})T = -\mathcal{J}(T^* \mathcal{J} T) = -\mathcal{J}^2 = I,$$

and $\|L\| \leq \|T\|$ because $\|\mathcal{J}\| = 1$. Finally $E^2 = T(LT)L = TL = E$ and $\text{ran } E = R$, which proves (iv). $\square$

The operator L is the adjoint of T with respect to Ω . Indeed,

$$\Omega(Lx, y) = g(\mathcal{J}Lx, y) = g(\mathcal{J}(-\mathcal{J} T^* \mathcal{J})x, y) = g(T^* \mathcal{J} x, y) = g(\mathcal{J}x, Ty) = \Omega(x, Ty).$$

Consequently

$$N := \ker L = \{v \in V : \Omega(v, Tx) = 0 \text{ for all } x \in V\} = R^{\perp \Omega}.$$

Lemma 2.2. Let T be a symplectic embedding, $R = \text{ran } T$, and $N = R^{\perp \Omega}$. Then:

- (i) $N = \mathcal{J}^{-1}(R^{\perp g})$ is closed, and $V = R \oplus N$ as a topological direct sum with $\Omega(R, N) = 0$;
- (ii) both $(R, \Omega|_R)$ and $(N, \Omega|_N)$ are strong symplectic Hilbert spaces;
- (iii) $T : V \rightarrow R$ is a symplectic isomorphism;
- (iv) T is surjective if and only if $N = \{0\}$.

Proof. The identity $N = \mathcal{J}^{-1}(R^{\perp_g})$ is immediate from $\Omega(v, r) = g(\mathcal{J}v, r)$. Let P_R be the orthogonal projection onto R and set $B := P_R \mathcal{J}|_R : R \rightarrow R$. Then $g(Br, s) = \Omega(r, s)$ for $r, s \in R$. We temporarily assume the nondegeneracy assertion in (ii) for R (proved independently in the next paragraph via the isomorphism $T : V \rightarrow R$), so B is bijective. The operator $\tilde{E} := B^{-1}P_R \mathcal{J}$ is then a bounded idempotent with range R and kernel N , whence $V = R \oplus N$ topologically. Comparing kernels and ranges with Lemma 2.1(iv) shows $\tilde{E} = E = TL$. By definition of N one has $\Omega(R, N) = 0$.

To see that $\Omega|_R$ is strong, observe that $T : V \rightarrow R$ is a linear bijection preserving Ω (Lemma 2.1 and (3)), and Ω is strong on V , so the transported form on R is strong. In particular B is an isomorphism, as used above.

Now suppose $n \in N$ and $\Omega(n, N) = 0$. Then also $\Omega(n, R) = 0$, so $\Omega(n, V) = 0$, hence $n = 0$. Thus $\Omega|_N$ is nondegenerate. Given $\phi \in N^*$, the functional $\tilde{\phi} := \phi \circ F$ lies in V^* . Choose $x \in V$ with $\flat(x) = \tilde{\phi}$, and write $x = r + n$ according to $V = R \oplus N$. For every $m \in N$,

$$\phi(m) = \Omega(r + n, m) = \Omega(n, m),$$

so $\flat_N(n) = \phi$. Therefore $\flat_N$ is bijective, and the open mapping theorem makes it a topological isomorphism.

Assertion (iii) is Lemma 2.1 together with (3), and (iv) is the equivalence $R = V \iff N = \{0\}$. $\square$

Lemma 2.3. *If T is an isometric symplectic embedding, then $N = R^{\perp_g}$. In particular the decomposition $V = R \oplus N$ is orthogonal, and $\mathcal{J}$ leaves both R and N invariant.*

Proof. From $T^*T = I$ and $T^*\mathcal{J}T = \mathcal{J}$ one obtains $T^* = -\mathcal{J}T^*\mathcal{J} = L$. Taking adjoints and using $\mathcal{J}^* = -\mathcal{J}$ yields $T = -\mathcal{J}T\mathcal{J}$, or equivalently $\mathcal{J}T = -T\mathcal{J}$. Hence

$$\mathcal{J}(R) = \mathcal{J}T(V) = T\mathcal{J}(V) = R,$$

and therefore $\mathcal{J}(R^{\perp_g}) = R^{\perp_g}$ as well. It follows that

$$R^{\perp_\Omega} = \mathcal{J}^{-1}(R^{\perp_g}) = R^{\perp_g}.$$

$\square$

3 Extension of a single symplectic embedding

Theorem 3.1. *Let $T : V \rightarrow V$ be a symplectic embedding. Then there exist a symplectic Hilbert space $(W, \hat{\Omega})$ containing (V, Ω) as a closed symplectic subspace and a symplectic automorphism $\tilde{T} : W \rightarrow W$ such that $\tilde{T}|_V = T$.*

Proof. If $N = \{0\}$, then T is already a symplectic automorphism by Lemma 2.2(iv), and there is nothing to prove. Assume henceforth that $N \neq \{0\}$.

For each integer $k \geq 1$ let (N_k, g_k, Ω_k) be a copy of $(N, g|_N, \Omega|_N)$, and fix a linear isomorphism $\iota_k : N \rightarrow N_k$ that is unitary for the Hilbert structures and satisfies $\Omega_k(\iota_k n, \iota_k n') = \Omega(n, n')$ for all $n, n' \in N$. Define

$$W := V \oplus \bigoplus_{k \geq 1} N_k \tag{4}$$

as an orthogonal direct sum of Hilbert spaces: its elements are sequences $(v; n_1, n_2, \dots)$ with $\|v\|^2 + \sum_{k \geq 1} \|n_k\|^2 < \infty$. Equip W with the continuous bilinear form

$$\hat{\Omega}((v; n_\bullet), (v'; n'_\bullet)) := \Omega(v, v') + \sum_{k \geq 1} \Omega(n_k, n'_k). \tag{5}$$

Continuity of $\widehat{\Omega}$ follows from $|\Omega(a, b)| \leq \|a\| \|b\|$ and the Cauchy–Schwarz inequality on ℓ^2 . The associated map $\widehat{\flat} : W \rightarrow W^*$ is the orthogonal direct sum $\flat \oplus \bigoplus_k \flat_N$. Each summand is a topological isomorphism (Lemma 2.2(ii)), and the operator norms of the inverses are uniformly bounded (each copy of N carries the same symplectic structure). Hence $\widehat{\flat}$ is a topological isomorphism, so $\widehat{\Omega}$ is strong. The isometric embedding $v \mapsto (v; 0, 0, \dots)$ realizes V as a closed symplectic subspace of W .

Define $\widetilde{T} : W \rightarrow W$ by

$$\widetilde{T}(v; n_1, n_2, n_3, \dots) = (Tv + \iota_1^{-1}n_1; \iota_1\iota_2^{-1}n_2, \iota_2\iota_3^{-1}n_3, \dots). \quad (6)$$

Each factor $\iota_k\iota_{k+1}^{-1}$ is unitary, and ι_1^{-1} is unitary, so

$$\begin{aligned} \|\widetilde{T}u\|^2 &= \|Tv + \iota_1^{-1}n_1\|^2 + \sum_{k \geq 1} \|n_{k+1}\|^2 \\ &\leq 2\|T\|^2\|v\|^2 + 2\|n_1\|^2 + \sum_{k \geq 1} \|n_{k+1}\|^2 \leq (2\|T\|^2 + 2)\|u\|^2. \end{aligned}$$

Thus $\widetilde{T} \in \mathcal{B}(W)$. If $n_\bullet = 0$, then $\widetilde{T}(v; 0) = (Tv; 0, \dots)$, so $\widetilde{T}|_V = T$.

We next verify that $\widetilde{T}$ preserves $\widehat{\Omega}$. Let $u = (v; n_\bullet)$ and $u' = (v'; n'_\bullet)$. Then

$$\widehat{\Omega}(\widetilde{T}u, \widetilde{T}u') = \Omega(Tv + \iota_1^{-1}n_1, Tv' + \iota_1^{-1}n'_1) + \sum_{k \geq 1} \Omega(n_{k+1}, n'_{k+1}).$$

Since $Tv \in R$, $\iota_1^{-1}n_1 \in N$, and $\Omega(R, N) = 0$, the cross terms vanish. Symplecticity of T and of ι_1 therefore yields

$$\Omega(Tv, Tv') = \Omega(v, v'), \quad \Omega(\iota_1^{-1}n_1, \iota_1^{-1}n'_1) = \Omega(n_1, n'_1).$$

Adding the shifted tail sum produces $\widehat{\Omega}(\widetilde{T}u, \widetilde{T}u') = \widehat{\Omega}(u, u')$.

Injectivity of $\widetilde{T}$ follows at once: if $\widetilde{T}u = 0$, then for every $u' \in W$

$$\widehat{\Omega}(u, u') = \widehat{\Omega}(\widetilde{T}u, \widetilde{T}u') = 0,$$

so $u = 0$ by strongness of $\widehat{\Omega}$.

For surjectivity, take an arbitrary $w = (z; m_1, m_2, \dots) \in W$. Set $n_{k+1} := \iota_{k+1}\iota_k^{-1}m_k$ for $k \geq 1$; then $\sum \|n_{k+1}\|^2 = \sum \|m_k\|^2 < \infty$. Decompose $z = z_R + z_N$ according to $V = R \oplus N$, and put $v := T^{-1}z_R \in V$ together with $n_1 := \iota_1 z_N$. The vector $u = (v; n_1, n_2, \dots)$ belongs to W and satisfies $\widetilde{T}u = w$.

Thus $\widetilde{T}$ is a bounded bijective endomorphism of the Hilbert space W , so $\widetilde{T}^{-1}$ is bounded by the bounded inverse theorem. The inverse preserves $\widehat{\Omega}$ because $\widetilde{T}$ does. Therefore $\widetilde{T} \in \text{Sp}(W, \widehat{\Omega})$. $\square$

Remark 3.2. Writing $E = TL$ and $F = I - E$, the inverse is given explicitly by

$$\widetilde{T}^{-1}(z; m_1, m_2, \dots) = (Lz; \iota_1 Fz, \iota_2 \iota_1^{-1}m_1, \iota_3 \iota_2^{-1}m_2, \dots). \quad (7)$$

Indeed, a direct substitution into (6) recovers the identity on W , using $TLz + Fz = z$ and $\iota_k \iota_{k+1}^{-1} \iota_{k+1} \iota_k^{-1} = \text{id}$. Iterating on vectors of V yields, for every $k \geq 1$ and $v \in V$,

$$\widetilde{T}^{-k}v = (L^k v; \iota_1 F L^{k-1} v, \dots, \iota_k F v, 0, 0, \dots). \quad (8)$$

In particular, if $n \in N$ then $Ln = 0$ and $Fn = n$, so

$$\widetilde{T}^{-k}n = (0; 0, \dots, 0, \iota_k n, 0, \dots). \quad (9)$$

Remark 3.3. If T is an isometric symplectic embedding, then the operator $\tilde{T}$ of Theorem 3.1 is unitary. Indeed, Lemma 2.3 provides an orthogonal decomposition $V = R \oplus N$ with $\|Tv\| = \|v\|$ and with $\iota_1^{-1}n_1 \perp Tv$. Consequently

$$\|\tilde{T}u\|^2 = \|Tv\|^2 + \|\iota_1^{-1}n_1\|^2 + \sum_{k \geq 1} \|n_{k+1}\|^2 = \|v\|^2 + \sum_{k \geq 1} \|n_k\|^2 = \|u\|^2.$$

Combined with the surjectivity already proved, $\tilde{T}$ is unitary. Moreover, the same computation that gave $\mathcal{J}T = -T\mathcal{J}$ extends componentwise to show that $\tilde{T}$ anticommutes with the block-diagonal extension of $\mathcal{J}$ to W , so $\tilde{T}$ is orthosymplectic.

Example 3.4. If T is the unilateral block shift on $\bigoplus_{n \geq 1} \mathbb{R}^{2d}$, then N is the first summand and $\tilde{T}$ is unitarily equivalent to the bilateral block shift on $\bigoplus_{n \in \mathbb{Z}} \mathbb{R}^{2d}$.

4 Extension of a commuting isometric pair

Throughout this section, T and T' denote *isometric* symplectic embeddings of V into itself that commute: $TT' = T'T$. We do *not* assume that they doubly commute (i.e., we do not assume $T^*T' = T'T^*$). Commuting isometries need not doubly commute in general, and the argument below nowhere requires this stronger relation.

Theorem 4.1. *Let $T, T' : V \rightarrow V$ be commuting isometric symplectic embeddings. Then there exist a symplectic Hilbert space $(W, \hat{\Omega})$ containing (V, Ω) as a closed symplectic subspace and commuting unitary symplectic automorphisms $\tilde{T}, \tilde{T}'$ of W such that $\tilde{T}|_V = T$ and $\tilde{T}'|_V = T'$.*

Proof. By Theorem 3.1 and Remark 3.3 there exist a symplectic Hilbert space $(W_1, \hat{\Omega}_1)$ and a unitary symplectic automorphism U of W_1 with $U|_V = T$. Replacing W_1 by the closed subspace

$$W_1 = \overline{\text{span}}\{U^k v : k \in \mathbb{Z}, v \in V\} \quad (10)$$

if necessary, we may assume that the extension is minimal; the subspace (10) reduces U and inherits a strong symplectic form from $\hat{\Omega}_1$.

Construction of a densely defined intertwiner. Let $\mathcal{D}_0 := \text{span}\{U^k v : k \in \mathbb{Z}, v \in V\}$, which is dense in W_1 by minimality. Define a linear map $W_0 : \mathcal{D}_0 \rightarrow W_1$ on generators by

$$W_0(U^k v) := U^k(T'v), \quad k \in \mathbb{Z}, v \in V, \quad (11)$$

and extend by linearity. To see that W_0 is well-defined and isometric, it suffices to prove the inner-product identity

$$g(U^k(T'x), U^\ell(T'y)) = g(U^k x, U^\ell y) \quad (12)$$

for all $k, \ell \in \mathbb{Z}$ and all $x, y \in V$. Indeed, if a linear combination $\xi = \sum_i U^{k_i} x_i$ vanishes, then

$$\left\| W_0 \xi \right\|^2 = \sum_{i,j} g(U^{k_i}(T'x_i), U^{k_j}(T'x_j)) = \sum_{i,j} g(U^{k_i} x_i, U^{k_j} x_j) = \|\xi\|^2 = 0,$$

so $W_0 \xi = 0$; thus W_0 is independent of the representation of its argument. The same computation shows $\|W_0 \xi\| = \|\xi\|$ on $\mathcal{D}_0$.

Since U is unitary, both sides of (12) depend only on the difference $k - \ell$. Replacing (k, ℓ) by $(k - \ell, 0)$ when $k \geq \ell$, and using unitarity to pass from the case $k < \ell$ to the case $k > \ell$ upon swapping the two arguments, one reduces (12) to the family of identities

$$g(U^m(T'x), T'y) = g(U^m x, y), \quad (13)$$

$$g(T'x, U^m(T'y)) = g(x, U^m y), \quad (14)$$

for all integers $m \geq 0$ and all $x, y \in V$. Identity (14) is the same as (13) with the roles of the two arguments interchanged, so it is enough to prove (13).

For $m = 0$, (13) is the isometry relation $g(T'x, T'y) = g(x, y)$. Now fix $m \geq 1$. On the original space V one has $TT' = T'T$ and $U|_V = T$, whence by a trivial induction

$$U^m(T'v) = T'(U^mv) \quad \text{for all } v \in V \text{ and all } m \geq 0. \quad (15)$$

(Negative powers of U are not needed on V at this stage.) Consequently

$$g(U^m(T'x), T'y) = g(T'(U^mx), T'y) = g(U^mx, y),$$

where the second equality uses that T' is isometric. This is (13).

Thus W_0 is a well-defined isometric linear map on the dense subspace $\mathcal{D}_0$. It extends uniquely by uniform continuity to an isometry $W \in \mathcal{B}(W_1)$ of W_1 into itself.

Extension and intertwining properties. For every $v \in V$ one has $Wv = W_0v = T'v$, so $W|_V = T'$. Moreover, if $U^k v \in \mathcal{D}_0$, then

$$WU(U^k v) = W(U^{k+1}v) = U^{k+1}(T'v) = U(U^k(T'v)) = UW(U^k v).$$

By denseness of $\mathcal{D}_0$ and continuity, $WU = UW$.

Preservation of the symplectic form. Write $\widehat{\Omega}$ for $\widehat{\Omega}_1$. Let $k, \ell \in \mathbb{Z}$ and $x, y \in V$. Then

$$\widehat{\Omega}(WU^k x, WU^\ell y) = \widehat{\Omega}(U^k T'x, U^\ell T'y).$$

Since U preserves $\widehat{\Omega}$,

$$\widehat{\Omega}(U^k T'x, U^\ell T'y) = \widehat{\Omega}(T'x, U^{\ell-k} T'y).$$

Set $m := \ell - k$. If $m \geq 0$, then U^m agrees with T^m on V , and (15) gives $U^m T'y = T' T^m y$. Therefore

$$\widehat{\Omega}(T'x, U^m T'y) = \Omega(T'x, T' T^m y) = \Omega(x, T^m y) = \widehat{\Omega}(x, U^m y) = \widehat{\Omega}(U^k x, U^\ell y),$$

where the second equality uses that T' preserves Ω . If $m = -p$ with $p > 0$, unitarity of U moves the negative power to the first argument:

$$\begin{aligned} \widehat{\Omega}(T'x, U^{-p} T'y) &= \widehat{\Omega}(U^p T'x, T'y) = \widehat{\Omega}(T' T^p x, T'y) = \Omega(T^p x, y) \\ &= \widehat{\Omega}(U^p x, y) = \widehat{\Omega}(x, U^{-p} y) = \widehat{\Omega}(U^k x, U^\ell y). \end{aligned}$$

Thus $\widehat{\Omega}(W\xi, W\eta) = \widehat{\Omega}(\xi, \eta)$ for all $\xi, \eta \in \mathcal{D}_0$. By continuity, W preserves $\widehat{\Omega}$ on all of W_1 . In other words, W is an isometric symplectic embedding of W_1 into itself, and W commutes with the unitary symplectic automorphism U .

Unitary dilation of W . Apply Theorem 3.1 in the isometric situation of Remark 3.3 to the operator W on W_1 . One obtains a symplectic Hilbert space $(W, \widehat{\Omega})$ containing W_1 and a unitary symplectic automorphism $\widetilde{T}'$ of W with $\widetilde{T}'|_{W_1} = W$. Replacing W by the closed span of $\{(\widetilde{T}')^k w_1 : k \in \mathbb{Z}, w_1 \in W_1\}$ if needed, we may assume that this second dilation is likewise minimal.

Extension of U by conjugation. Let $\mathcal{D}_1 := \text{span}\{(\widetilde{T}')^k w_1 : k \in \mathbb{Z}, w_1 \in W_1\}$, dense in W by the minimality just arranged. Define $U_0 : \mathcal{D}_1 \rightarrow W$ on generators by

$$U_0((\widetilde{T}')^k w_1) := (\widetilde{T}')^k (U w_1), \quad k \in \mathbb{Z}, w_1 \in W_1, \quad (16)$$

and extend linearly. The same reasoning as for W_0 shows that U_0 is well-defined and isometric, once one verifies

$$g((\widetilde{T}')^k (U\xi), (\widetilde{T}')^\ell (U\eta)) = g((\widetilde{T}')^k \xi, (\widetilde{T}')^\ell \eta) \quad (17)$$

for $k, \ell \in \mathbb{Z}$ and $\xi, \eta \in W_1$. Because $\tilde{T}'$ is unitary, it is enough to treat $\ell = 0$ and $k = m \geq 0$, together with the symmetric case obtained by swapping arguments. For $m = 0$ the claim is that U is isometric on W_1 , which is true. For $m \geq 1$ we use that $\tilde{T}'|_{W_1} = U$ commutes with U , whence by induction

$$(\tilde{T}')^m(U\xi) = U((\tilde{T}')^m\xi) \quad \text{for all } \xi \in W_1 \text{ and } m \geq 0.$$

Therefore

$$g((\tilde{T}')^m(U\xi), U\eta) = g(U((\tilde{T}')^m\xi), U\eta) = g((\tilde{T}')^m\xi, \eta),$$

as required. Extending by continuity, one obtains an isometry $\tilde{U} \in \mathcal{B}(W)$ satisfying $\tilde{U}|_{W_1} = U$ and $\tilde{U}\tilde{T}' = \tilde{T}'\tilde{U}$.

The identical computation as in the form-preservation paragraph above, with the pair (U, W) replaced by the pair $(\tilde{U}, \tilde{T}')$ and with W_1 playing the role formerly played by V , shows that $\tilde{U}$ preserves $\hat{\Omega}$. (Explicitly: for $m \geq 0$,

$$\hat{\Omega}((\tilde{T}')^m U\xi, U\eta) = \hat{\Omega}(U(\tilde{T}')^m \xi, U\eta) = \hat{\Omega}((\tilde{T}')^m \xi, \eta),$$

while for negative exponents one again moves powers with unitarity of $\tilde{T}'$.)

Surjectivity of $\tilde{U}$. The range of the isometry $\tilde{U}$ is a closed subspace of W . It contains W_1 because $\tilde{U}|_{W_1} = U$ is unitary on W_1 , hence surjective onto W_1 . Moreover $\text{ran } \tilde{U}$ is invariant under $\tilde{T}'$: if $w = \tilde{U}z$, then

$$\tilde{T}'w = \tilde{T}'\tilde{U}z = \tilde{U}(\tilde{T}'z) \in \text{ran } \tilde{U}.$$

Since $\tilde{T}'$ is unitary, $\text{ran } \tilde{U}$ is likewise invariant under $(\tilde{T}')^{-1}$. Therefore $\text{ran } \tilde{U}$ contains every vector $(\tilde{T}')^k w_1$ with $k \in \mathbb{Z}$ and $w_1 \in W_1$. By minimality of the dilation of W , one concludes that $\text{ran } \tilde{U} = W$. An isometric surjective operator on a Hilbert space is unitary, so $\tilde{U}$ is unitary.

Set $\tilde{T} := \tilde{U}$ and retain the notation $\tilde{T}'$ for the unitary already constructed. Then $\tilde{T}$ and $\tilde{T}'$ are commuting unitary symplectic automorphisms of W extending T and T' respectively. $\square$

Corollary 4.2. *The same conclusion holds for any finite or countably infinite family of mutually commuting isometric symplectic embeddings. One proceeds by induction on the number of operators (or by exhaustion in the countable case): at stage n , the first n operators have been extended to commuting unitary symplectic automorphisms of a common space $W^{(n)}$, and the next isometry is conjugated onto the minimal unitary dilation of one of them (or of a fixed amalgamation) exactly as in the proof of Theorem 4.1.*

Example 4.3. If T is the unilateral block shift on $\bigoplus_{n \geq 1} \mathbb{R}^{2d}$ and $T' = T^m$ for some integer $m \geq 1$, then both maps are isometric symplectic embeddings, they commute, and Theorem 4.1 recovers the bilateral block shift together with its m -th power on $\bigoplus_{n \in \mathbb{Z}} \mathbb{R}^{2d}$.

5 Remarks on the non-isometric commuting case

Theorem 3.1 requires no isometry assumption: an arbitrary bounded symplectic embedding extends to a symplectic automorphism. The conjugation method of Theorem 4.1, however, uses isometry of T' (and unitarity of the dilation of T) in an essential way. Two obstructions appear if one drops isometry.

First, if T and T' are commuting symplectic embeddings that are not isometric, the defect space $N = (\text{ran } T)^{\perp \Omega}$ need not be invariant under T' . A concrete illustration on $\bigoplus_{n \geq 1} \mathbb{R}^2$ is given by the commuting pair

$$\begin{aligned} T(x_1, x_2, x_3, \dots) &= (0, 2x_1, \tfrac{1}{2}x_2, 2x_3, \tfrac{1}{2}x_4, \dots), \\ T'(x_1, x_2, x_3, \dots) &= (0, 3x_1, \tfrac{1}{3}x_2, 3x_3, \tfrac{1}{3}x_4, \dots), \end{aligned}$$

each of which is a (non-surjective, non-isometric) symplectic embedding for the orthogonal sum of area forms. Here N is the first summand, while $T'N$ lies in the second summand and meets N only at 0. Consequently one cannot define a fibrewise extension of T' on the defect slots of the construction in Theorem 3.1.

Second, if one nevertheless forms the minimal automorphism extension $U = \tilde{T}$ of T furnished by Theorem 3.1 and attempts to define a densely defined intertwiner by the same formula $U^k v \mapsto U^k(T'v)$, the inner-product identities that produced an isometric (hence bounded) map on the dense subspace fail: one no longer has $\|U^k(T'x)\| = \|U^k x\|$. Boundedness of the conjugated operator on the whole minimal dilation may therefore fail.

A joint commuting extension to symplectic automorphisms is nevertheless expected on general grounds after complexification, by viewing T and T' as commuting J -norm-preserving operators on a Kreĭn space and invoking the corresponding dilation theory [3]. An alternative elementary approach would be to pass to a carefully chosen non-minimal enlargement on which a bounded intertwiner can be arranged; we do not develop that construction here. The isometric case treated in Theorem 4.1 is the natural symplectic analogue of Itô's theorem and is the setting in which the conjugation method closes in a fully elementary way.

References

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